

Day 1: HTML Intro & Basic Structure

Web Development Fundamentals — Instructor & Student Handbook

DAY 1 CURRICULUM

1. WHAT IS HTML?

Definition

HTML stands for HyperText Markup Language. It is used to create the core **structure and content** of a webpage. It is a markup language, not a programming language.

Real-Life Analogy: Building a House 🏠

1. HTML (Structure)

Acts as the brick wall, concrete columns, and timber framework of the house.

2. CSS (Design)

Acts as the wall paint, ceramic tiles, interior decor, and overall visual appeal.

3. JavaScript (Action)

Acts as the electrical switches, automated doors, and smart appliances.

HTML uses **tags** to instruct web browsers on how to render elements on screen:

```
<h1>Welcome to My Website</h1> <p>This is my first webpage.</p>
```

Anatomy of an HTML Element:

Opening Tag

<p>

+

Content

Hello World

+

Closing Tag

</p>

=

HTML Element

<p>Hello World</p>

2. BASIC HTML STRUCTURE

Every standard HTML document follows a strict foundational skeleton:

```
<!-- Tells browser this is an HTML5 document --> <!DOCTYPE html> <html> <head> <title>My First Website</title> </head> <body> <h1>Welcome to My Website</h1> <p>This is my first web page.</p> </body> </html>
```

Component	Purpose & Description
<!DOCTYPE html>	Mandatory declaration specifying that the file is an HTML5 document.
<html>	The root element that wraps all content on the entire webpage.
<head>	Contains metadata, search engine information, external stylesheets, and page titles.
<title>	Defines the title displayed on the browser's title tab.
<body>	Encloses all user-visible content (headings, paragraphs, images, links).

3. HEADINGS (`<H1>` TO `<H6>`)

HTML provides six levels of headings. Headings form the structural document outline.

```
<h1>Main Heading (Level 1)</h1> <h2>Section Heading (Level 2)</h2> <h3>Sub-section Heading (Level 3)</h3> <h4>Heading 4</h4> <h5>Heading 5</h5> <h6>Heading 6 (Smallest)</h6>
```

Core Rule: Document Hierarchy Over Font Sizing

Do not use heading tags simply to make text bigger or bolder. Use them to maintain a logical sequence (e.g., an `<h2>` nested beneath an `<h1>`).

Example of Proper Hierarchy:

```
<h1>Web Development Course</h1> <h2>HTML Basics</h2> <p>HTML creates webpage structure.</p> <h2>CSS Styling</h2> <p>CSS adds visual styling and presentation.</p>
```

4. PARAGRAPHS (`<P>`)

Paragraphs are defined with the `<p>` tag. Browsers automatically append vertical spacing before and after paragraphs.

```
<p>HTML is simple to learn and essential for building websites.</p> <p>CSS handles typography, colors, and layout positioning.</p>
```

Browser Whitespace Handling

HTML collapses multiple spaces and line breaks in code into a single space on screen. Writing text across multiple lines inside a `<p>` tag will still render as a single continuous line unless tags like `<br>` are used.

5. LINKS (`<A>`)

Hyperlinks are created using the anchor element (`<a>`). The `href` attribute defines the destination URL.

```
<a href="https://www.google.com">Visit Google</a>
```

Opening links in a new browser tab: Use the `target="_blank"` attribute.

```
<a href="https://www.youtube.com" target="_blank">Open YouTube in New Tab</a>
```

6. IMAGES (`<IMG>`)

Images use the self-closing `<img>` element. They require two core attributes: `src` (path) and `alt` (fallback text).

```

```

7. COMBINED PRACTICAL STARTER (`INDEX.HTML`)

This single file combines all concepts taught in Day 1:

```
<!DOCTYPE html> <html> <head> <title>My First Webpage</title> </head> <body> <h1>Welcom
e to My Personal Website</h1> <h2>About Me</h2> <p>Hello! My name is Teja. I am learning web
development today.</p> <h2>My Favorite Website</h2> <p><a href="https://www.google.com" targe
t="_blank">Visit Google</a></p> <h2>My Favorite Photo</h2>  </body> </html>
```

8. CLASS SCHEDULE & TEACHING TIMELINE (120 MINS)

Time	Module	Classroom Action & Teaching Focus
00 - 05 min	Opening Question	Ask: "How does the browser know what is a heading, paragraph, image or link?"
05 - 15 min	HTML Overview	Explain full form, purpose, markup language concept, and House Analogy.
15 - 35 min	Live Coding Setup	Open VS Code, create index.html, write boilerplate, preview in Chrome.
35 - 55 min	Headings & Paragraphs	Guide students to test <h1>-<h6> and paragraph line collapsing.
55 - 70 min	Links (<a>)	Demonstrate linking to Google/YouTube and using target="_blank".
70 - 85 min	Images ()	Guide image placement in folder, testing src paths and alt tags.
85 - 120 min	Practical Lab	Mini-Project: Students build their own "My Profile Webpage".

9. HANDS-ON LAB EXERCISE: "MY PROFILE WEBPAGE"

Have students independently construct a personal profile document adhering to these requirements:

DESIRED STUDENT PAGE OUTPUT

MY PROFILE PAGE

[Image / Photo: my-photo.jpg]

About Me

Hello! My name is [Student Name]. I am learning Web Development.

My Education

Currently pursuing [Degree / Diploma / School Name].

Favorite Website

Visit Google (Opens in new tab)